

Trader Behavior Insights – Final Report

1. Introduction

This project explores the relationship between Bitcoin market sentiment (Fear–Greed Index) and trader performance using historical Hyperliquid trading data.

The goal is to understand how emotions in the market (fear or greed) influence profitability, risk-taking, and trade outcomes.

2. Datasets Used

1) Historical Trader Data (Hyperliquid)

Contains:

- Timestamp (trade time)
- Account
- Coin
- Execution price
- Trade size
- Side (Buy/Sell)
- Closed PnL
- Leverage / Position size
- Event details

2) Bitcoin Fear–Greed Index

Contains:

- Date
- Classification (Fear, Neutral, Greed, Extreme Fear, Extreme Greed)
- Sentiment score

These two datasets were merged on the date column to study how daily market sentiment affects trader decisions.

3. Methodology

Step 1 — Data Cleaning

- Parsed timestamps from the trading dataset

- Converted sentiment dates into standard format
- Removed invalid or missing timestamps
- Normalized column names

Step 2 — Data Merging

- Created date column from timestamps
- Kept one sentiment value per day
- Performed left-join:
- `merged_data = trades ← sentiment (on date)`

Step 3 — Exploratory Data Analysis (EDA)

- Plotted PnL distributions under different sentiment states
- Generated time-series trends of daily average PnL
- Analyzed volatility and outliers
- Compared trader behavior in Fear vs Greed days

Step 4 — Trader Feature Engineering

For each trader (Account), calculated:

- Total trades
- Total PnL
- Average PnL
- Win rate (number of profitable trades / total trades)
- Standard deviation of PnL
- Average leverage (if available)

The output is stored in `trader_features.csv`.

4. Key Insights

1) Sentiment affects profitability

- Greed days often show higher volatility in PnL.
- Fear and Extreme Fear days show more consistent negative PnL for many traders.
- Boxplot shows wider spread on Greed days → higher risk-taking.

2) High leverage + Fear sentiment = higher losses

Trades with large leverage performed poorly especially during market Fear phases.

3) Some traders remain consistent

A few accounts have high win rates even in fear conditions, indicating disciplined and stable strategies.

4) Market sentiment score correlates with PnL variance

Higher sentiment scores (Greed) lead to more extreme PnL outcomes — both profits and losses.

5. Visual Outputs Included

- Closed PnL by Sentiment (boxplot)
 - Daily Average PnL over Time
- These figures are included in the outputs/ folder.

6. Recommendations

1) Reduce leverage during Fear / Extreme Fear days

Because traders tend to lose more when market sentiment is negative.

2) Use sentiment category as a risk control filter

A trading strategy can selectively reduce position size on Fear days.

3) Monitor sentiment-lag effects

Performance often shifts 1 day after a sentiment spike useful for systematic strategies.

7. Limitations

- Timestamp timezones may differ (IST vs UTC)
- Outliers in PnL can affect summary statistics
- Sentiment is daily intraday sentiment not included
- Dataset size may limit predictive modelling

8. Conclusion

This study shows a clear link between market sentiment and trader performance. Greed days create more opportunities but also more risk, while Fear days consistently impact profitability. The engineered trader features and EDA insights can help design better trading strategies and risk controls in real-world Web3 trading environments.

Prepared By:

Munna Das

Junior Data Scientist – Trader Behavior Insights (Assignment Submission)